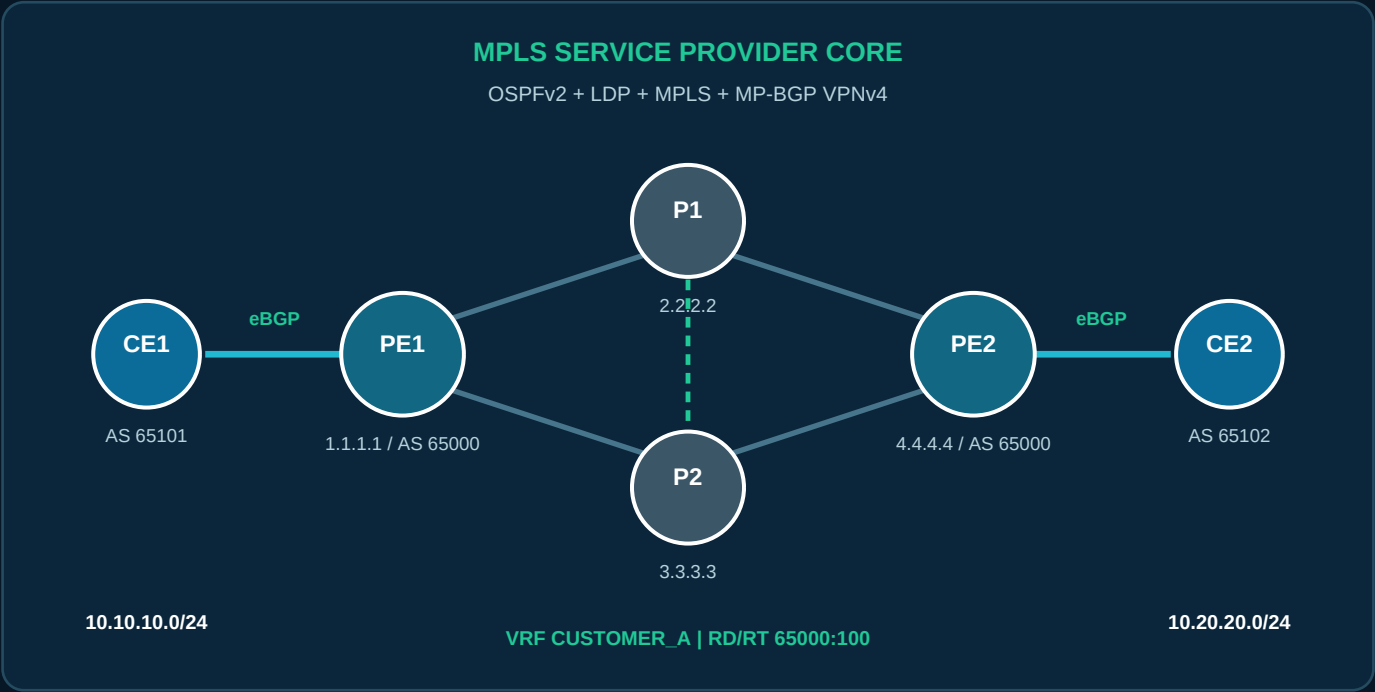


MPLS Layer-3 VPN

A service-provider-style lab using OSPFv2, LDP, MPLS, MP-BGP VPNv4, VRFs, and eBGP



6 ROUTERS	1 CUSTOMER VPN	4 CONTROL-PLANE TECHNOLOGIES
CE1 - PE1 - P1 - P2 - PE2 - CE2	VRF CUSTOMER_A	OSPF LDP MP-BGP eBGP

Prepared by Marc Lavoie
Technical portfolio report | August 2026

01 | EXECUTIVE OVERVIEW

A true service-provider VPN architecture

This lab separates customer routing, provider reachability, label distribution, and VPN route exchange into distinct protocol responsibilities. The result mirrors the operating model of a real MPLS Layer-3 VPN service rather than a collection of unrelated routing commands.

DESIGN OBJECTIVE

Provide private reachability between 10.10.10.0/24 and 10.20.20.0/24 while keeping customer prefixes out of the provider P-router routing tables.

Protocol responsibilities

Technology	Job in the architecture	Expected evidence
OSPFv2 Area 0	Provides provider underlay reachability between PE and P loopbacks.	Stable adjacencies and loopback routes.
MPLS LDP	Assigns and exchanges transport labels for provider prefixes.	LDP neighbors and LFIB entries.
MP-BGP VPNv4	Distributes customer VPN routes, route targets, and VPN labels between PE routers.	Established VPNv4 session and remote VPN routes.
VRF CUSTOMER_A	Maintains a separate customer routing table on each PE.	Customer routes appear only in the VRF.
eBGP	Exchanges customer routes across each CE-to-PE edge.	Local CE prefixes learned in the correct VRF.

Skills demonstrated

- Service-provider routing design and protocol-role separation.
- MPLS forwarding, LDP label distribution, and LFIB verification.
- VRF isolation, route distinguishers, and route-target policy.
- MP-BGP VPNv4 and CE-to-PE eBGP integration.
- Structured validation using control-plane and data-plane evidence.

02 | ARCHITECTURE

Six routers, one customer VPN, two forwarding paths

The customer edge routers exchange routes only with their local provider edge. The provider core transports labeled packets between PE routers without learning customer IPv4 prefixes.

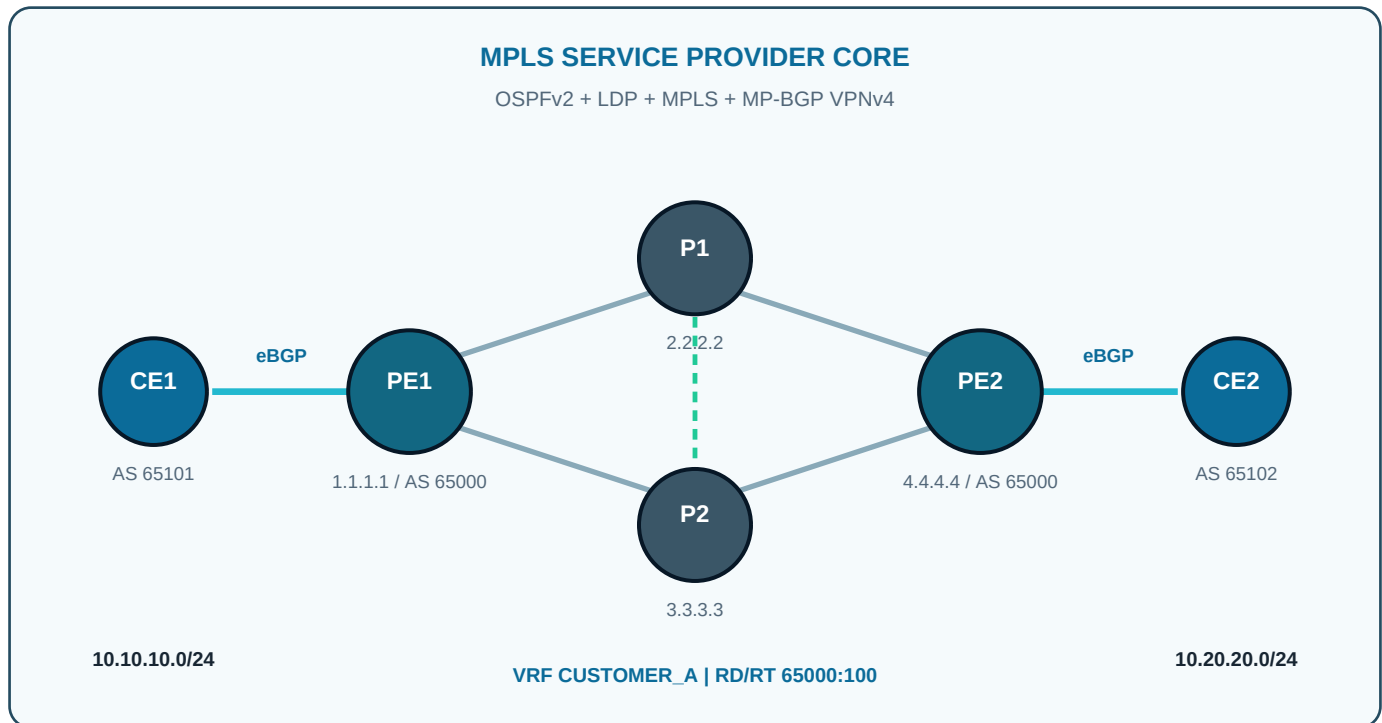


Figure 1. Logical topology and protocol boundaries.

CUSTOMER ROUTING

CE1 <-> PE1 and CE2 <-> PE2 use eBGP. Customer routes enter VRF CUSTOMER_A at the provider edge.

PROVIDER UNDERLAY

PE1, P1, P2, and PE2 use OSPF Area 0 and LDP. MPLS transports packets across either core path.

CORE DESIGN PRINCIPLE

P1 and P2 know only provider infrastructure. They should not contain 10.10.10.0/24 or 10.20.20.0/24 in the global routing table; they switch labeled packets toward the egress PE.

Expected P-router routes

```
P1 ROUTING TABLE (CONCEPTUAL)
0 1.1.1.1/32    PE1 loopback
0 3.3.3.3/32    P2 loopback
0 4.4.4.4/32    PE2 loopback
0 10.0.11.0/30   PE1-P1
0 10.0.13.0/30   P1-PE2
0 10.0.15.0/30   P1-P2
```

NOT EXPECTED: 10.10.10.0/24 or 10.20.20.0/24

03 | ADDRESSING PLAN

Deterministic addressing for fast troubleshooting

Loopbacks act as stable router identities and protocol endpoints. /30 core links make interface ownership and path tracing unambiguous.

Provider loopbacks

Router	Role	Loopback0	Use
PE1	Provider edge	1.1.1.1/32	LDP router ID and MP-BGP source
P1	Provider core	2.2.2.2/32	LDP router ID
P2	Provider core	3.3.3.3/32	LDP router ID
PE2	Provider edge	4.4.4.4/32	LDP router ID and MP-BGP source

MPLS core links

Link	Subnet	Endpoint A	Endpoint B
PE1 - P1	10.0.11.0/30	PE1 10.0.11.1	P1 10.0.11.2
PE1 - P2	10.0.12.0/30	PE1 10.0.12.1	P2 10.0.12.2
P1 - PE2	10.0.13.0/30	P1 10.0.13.1	PE2 10.0.13.2
P2 - PE2	10.0.14.0/30	P2 10.0.14.1	PE2 10.0.14.2
P1 - P2	10.0.15.0/30	P1 10.0.15.1	P2 10.0.15.2

Customer edge

Connection	Transit subnet	Provider	Customer	Customer LAN
CE1 - PE1	172.16.1.0/30	PE1 172.16.1.1	CE1 172.16.1.2	10.10.10.0/24
PE2 - CE2	172.16.2.0/30	PE2 172.16.2.1	CE2 172.16.2.2	10.20.20.0/24

AUTONOMOUS SYSTEMS

Provider: AS 65000 | CE1: AS 65101 | CE2: AS 65102 | PE-to-PE: MP-iBGP | CE-to-PE: eBGP

04 | VPN CONTROL PLANE

VRF policy and MP-BGP route exchange

The PE routers preserve customer isolation while exchanging VPN routes through MP-BGP. Matching route targets make both sites members of the same logical VPN.

VRF definition on PE1 and PE2

```
ip vrf CUSTOMER_A
 rd 65000:100
 route-target export 65000:100
 route-target import 65000:100
```

WHAT EACH FIELD DOES

The route distinguisher makes overlapping customer prefixes unique in VPNv4. The route target controls membership: exported routes carry 65000:100, and the matching import policy accepts them into CUSTOMER_A.

Route lifecycle

- 1 CE1 advertises 10.10.10.0/24 to PE1 through eBGP.
- 2 PE1 installs the prefix in VRF CUSTOMER_A and converts it into a VPNv4 route.
- 3 PE1 attaches RD 65000:100, RT 65000:100, and a VPN label, then advertises it to PE2 over MP-iBGP.
- 4 PE2 imports the route because CUSTOMER_A accepts RT 65000:100.
- 5 PE2 advertises the IPv4 route to CE2 through the customer-facing eBGP session. The reverse path follows the same process.

PE-to-PE relationship

```
PE1 Loopback0 1.1.1.1
      ||
      || MP-BGP VPNv4 (AS 65000)
      ||
PE2 Loopback0 4.4.4.4

Underlay prerequisite: OSPF must provide loopback reachability.
```

Separation of concerns

- OSPF carries provider infrastructure only.
- MP-BGP carries customer VPN reachability only.
- VRFs decide which customer table receives each VPN route.
- P routers remain unaware of customer IPv4 prefixes.

05 | MPLS DATA PLANE

Two labels carry each customer packet

The provider core forwards traffic with a transport label while the egress PE uses a VPN label to identify the correct customer VRF.

Encapsulation layer	Purpose
Customer application traffic	Original payload generated at the customer site
Customer IPv4	10.10.10.0/24 to 10.20.20.0/24
MPLS VPN label	Identifies the destination VPN/VRF at the egress PE
MPLS transport label	Carries the packet across the provider label-switched path
Ethernet	Hop-by-hop Layer-2 framing

LDP activation

```
mpls label protocol ldp
mpls ldp router-id Loopback0 force

interface <provider-facing-interface>
 mpls ip
```

Conceptual forwarding table

Local label	Outgoing label	FEC / prefix	Meaning
16	18	2.2.2.2/32	Reach P1 loopback
17	20	3.3.3.3/32	Reach P2 loopback
18	22	4.4.4.4/32	Reach PE2 loopback

Label values are dynamically allocated and will vary between devices and rebuilds.

Packet path from Site 1 to Site 2

- 1 PE1 receives a customer IPv4 packet from CE1 in VRF CUSTOMER_A.
- 2 PE1 pushes the VPN label learned through MP-BGP and the transport label learned through LDP.
- 3 P1 or P2 swaps only the transport label and forwards toward PE2.
- 4 PE2 removes the remaining label, selects VRF CUSTOMER_A, and performs an IPv4 lookup.
- 5 PE2 forwards the packet to CE2; the return traffic uses the reciprocal label stack.

06 | IMPLEMENTATION WORKFLOW

Build in layers and prove each dependency

A staged build prevents higher-level protocol problems from hiding basic interface or underlay failures.

Stage	Build focus	Gate before continuing
1	Base interfaces and loopbacks	All directly connected links are up/up
2	OSPF Area 0	Every provider loopback is reachable
3	LDP and MPLS	Provider neighbors and labels are present
4	VRF CUSTOMER_A	CE-facing interfaces are VRF-bound
5	PE-to-PE MP-BGP	VPNv4 session is established
6	CE-to-PE eBGP	Customer LANs enter the local VRF
7	Route-target import/export	Remote customer routes appear in each VRF
8	End-to-end validation	Sourced ping and traceroute succeed

Recommended build sequence

- 1 Cable the six-router topology and assign every physical interface and loopback address.
- 2 Enable OSPF Area 0 only on provider-facing links and advertise all provider loopbacks.
- 3 Confirm full provider loopback reachability before enabling MPLS.
- 4 Enable LDP globally, force Loopback0 as the LDP router ID, and apply mpls ip to provider interfaces.
- 5 Create VRF CUSTOMER_A on PE1 and PE2; bind each CE-facing interface before assigning its IP address.
- 6 Establish the PE1-to-PE2 MP-iBGP VPNv4 session using loopback sources and activate extended communities.
- 7 Configure CE-to-PE eBGP and advertise each customer LAN.
- 8 Validate route import/export, MPLS forwarding, and end-to-end customer reachability.

TROUBLESHOOTING RULE

Do not troubleshoot VPNv4 or VRF routing until OSPF and LDP are healthy. Each layer depends on the one below it: interfaces -> OSPF -> LDP/MPLS -> VRF -> MP-BGP -> eBGP -> application traffic.

Change-control notes

- Record the expected neighbor count before testing failure scenarios.
- Capture routing, label, and VPN tables at each stage as a known-good baseline.
- Use loopback-sourced tests so validation follows the same identities used by LDP and MP-BGP.
- Document dynamic labels as observations, not fixed design values.

07 | VERIFICATION RUNBOOK

Collect control-plane and data-plane evidence

The lab is complete only when protocol state, route separation, label forwarding, and customer reachability all agree.

Domain	Commands
OSPF underlay	show ip ospf neighbor show ip route ospf
MPLS and LDP	show mpls interfaces show mpls ldp neighbor show mpls forwarding-table
MP-BGP VPNv4	show bgp vpnv4 unicast all summary show bgp vpnv4 unicast all
VRF routing	show ip vrf show ip route vrf CUSTOMER_A show bgp ipv4 unicast vrf CUSTOMER_A
CE eBGP	show ip bgp summary
End-to-end	ping 10.20.20.1 source 10.10.10.1 traceroute 10.20.20.1 source 10.10.10.1

Final bidirectional tests

```
CE1# ping 10.20.20.1 source 10.10.10.1
CE2# ping 10.10.10.1 source 10.20.20.1
```

Evidence checklist

- OSPF adjacencies are FULL and every provider loopback is reachable.
- LDP neighbors match active MPLS links and transport labels exist for remote loopbacks.
- The VPNv4 session between 1.1.1.1 and 4.4.4.4 is Established.
- Both customer prefixes appear in VRF CUSTOMER_A on both PE routers.
- Neither customer prefix appears in a P-router global routing table.
- Bidirectional sourced pings succeed and traceroute crosses the expected provider path.

SUCCESS CONDITION

When CE1 and CE2 can reach each other's LAN interfaces while P1 and P2 remain customer-route-free, the lab has demonstrated a working MPLS Layer-3 VPN.

08 | OPERATIONAL ANALYSIS

Failure isolation and next-level extensions

A strong lab also proves that the design is observable, resilient to investigation, and ready to evolve beyond a single VPN.

Fault-isolation matrix

Symptom	Likely layer	First evidence to inspect
PE loopbacks unreachable	Interfaces / OSPF	Interface state, OSPF neighbors, OSPF routes
OSPF works but no labels	LDP / MPLS	MPLS-enabled interfaces and LDP neighbors
VPNv4 session down	MP-BGP	Loopback reachability, update source, remote AS, address-family activation
Remote route missing in VRF	Route-target policy	VPNv4 route, extended community, import/export RT
Routes exist but ping fails	Data plane / CE return path	LFIB, VRF CEF entry, CE routes, sourced traceroute

Expansion roadmap

Enhancement	Engineering value
Dual-homed customer sites	Tests redundancy, path preference, and edge failure behavior.
Multiple VRFs	Demonstrates tenant isolation and overlapping address support.
Route reflectors	Scales VPNv4 control-plane relationships beyond a full mesh.
BFD and provider failures	Measures failure detection and convergence behavior.
BGP communities and QoS	Adds policy control and service differentiation.
MPLS traffic engineering	Introduces explicit path control and bandwidth-aware transport.
Alternate CE protocol	Tests PE-CE integration with OSPF or EIGRP at one site.

PORTFOLIO TAKEAWAY

This project demonstrates the ability to design, build, validate, and explain a multi-protocol provider network - not merely enter isolated router commands.

Prepared for Marc Lavoie's public Network Engineering portfolio.